

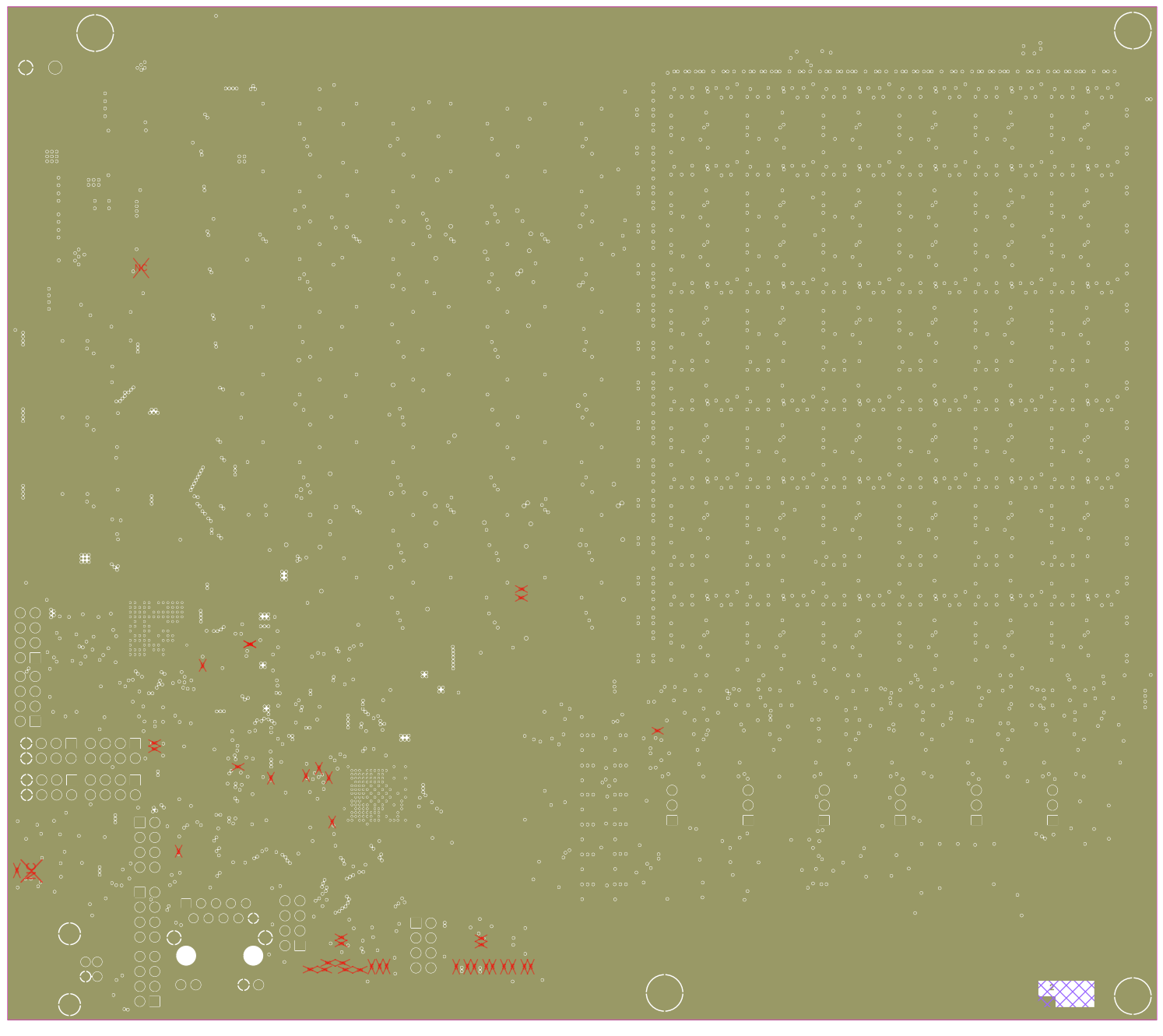
Proof-of-Concept PCB for Future ASIC | Patent Pending: SE 2630397-4

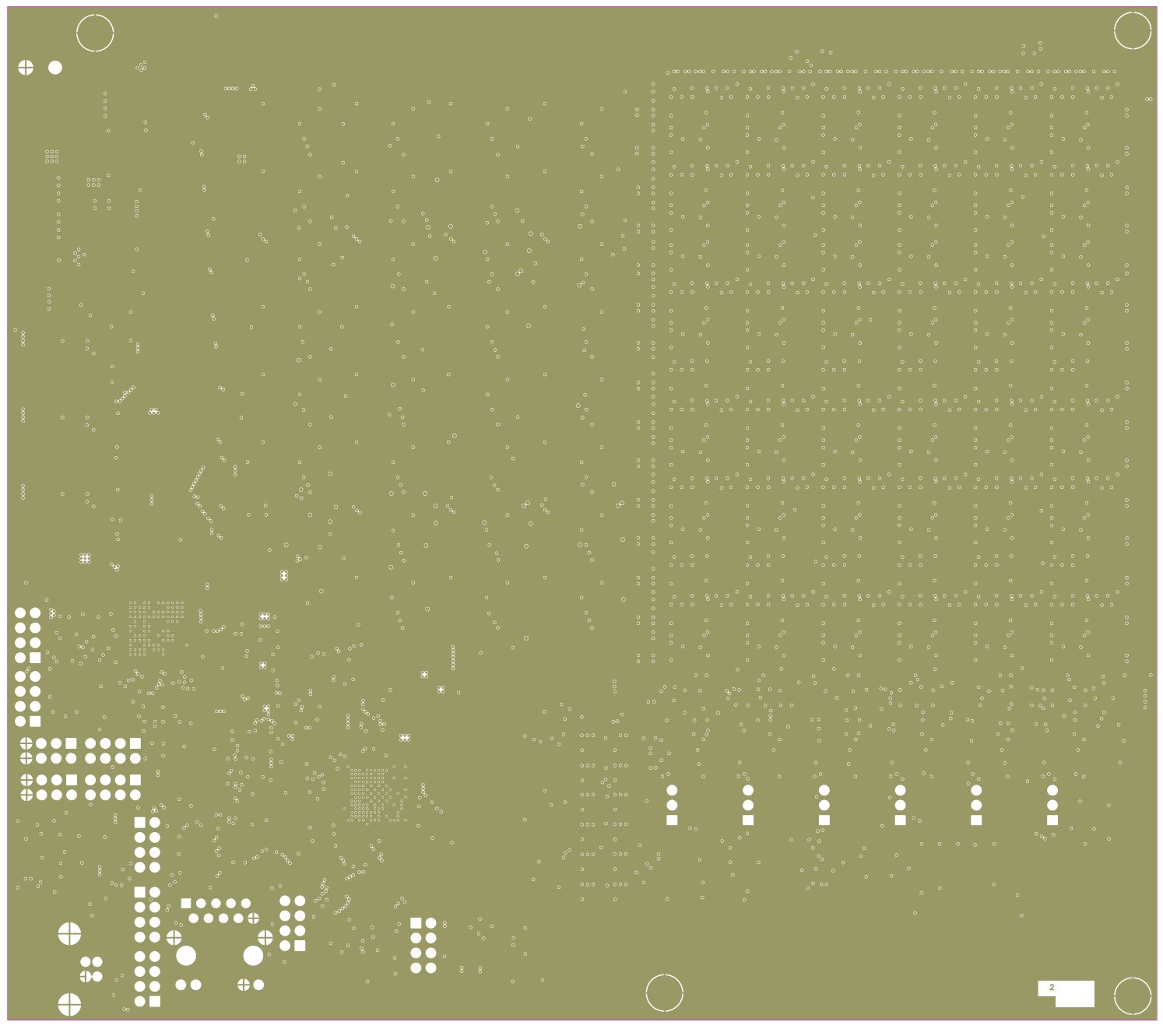
Source code and documentation link:
https://github.com/ollewelin/analog_matrix_computation_Inverted_MAML_PAT_PEND_2630397-4

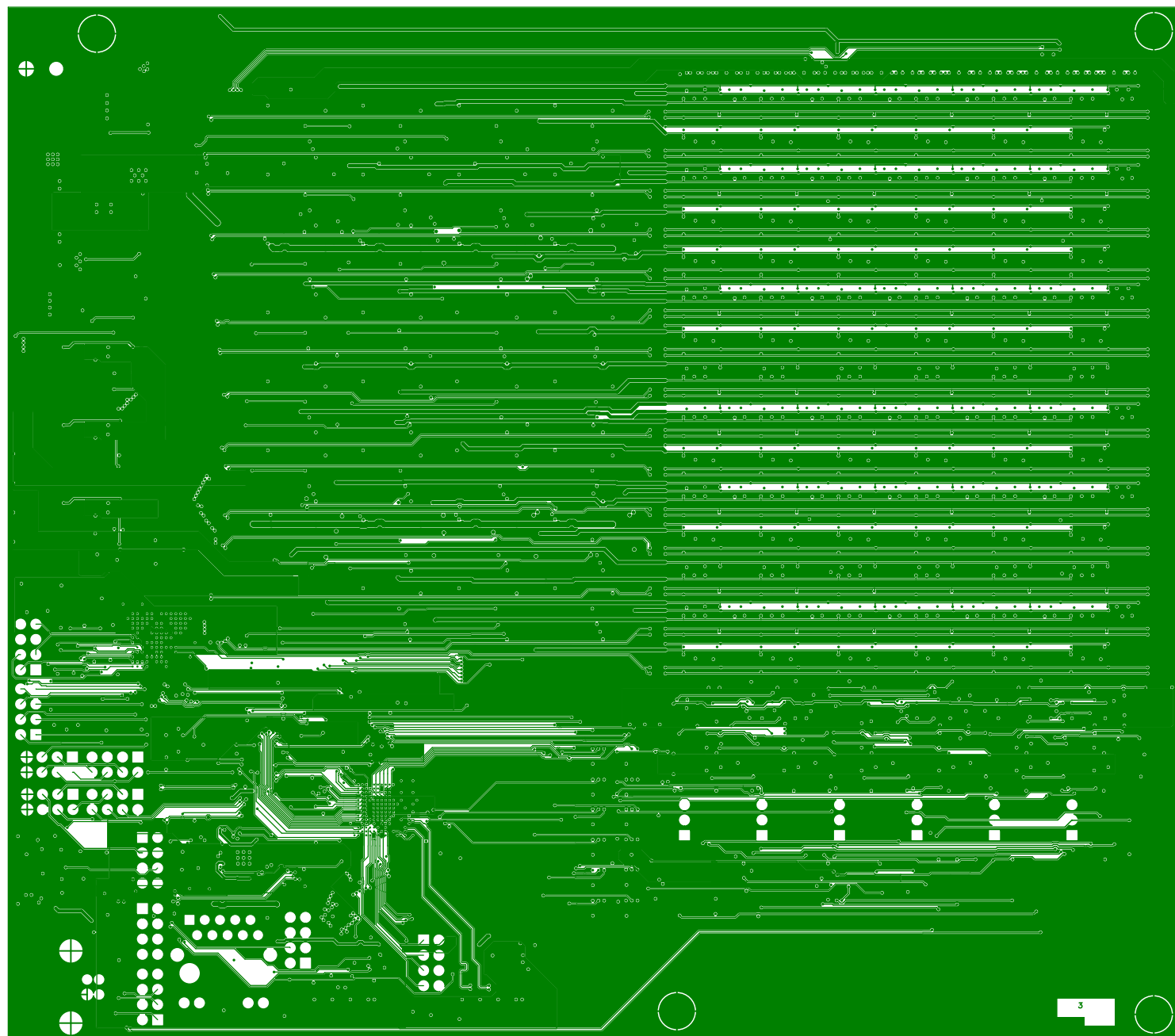
Inverted_MAML_IMC version 1.3.1

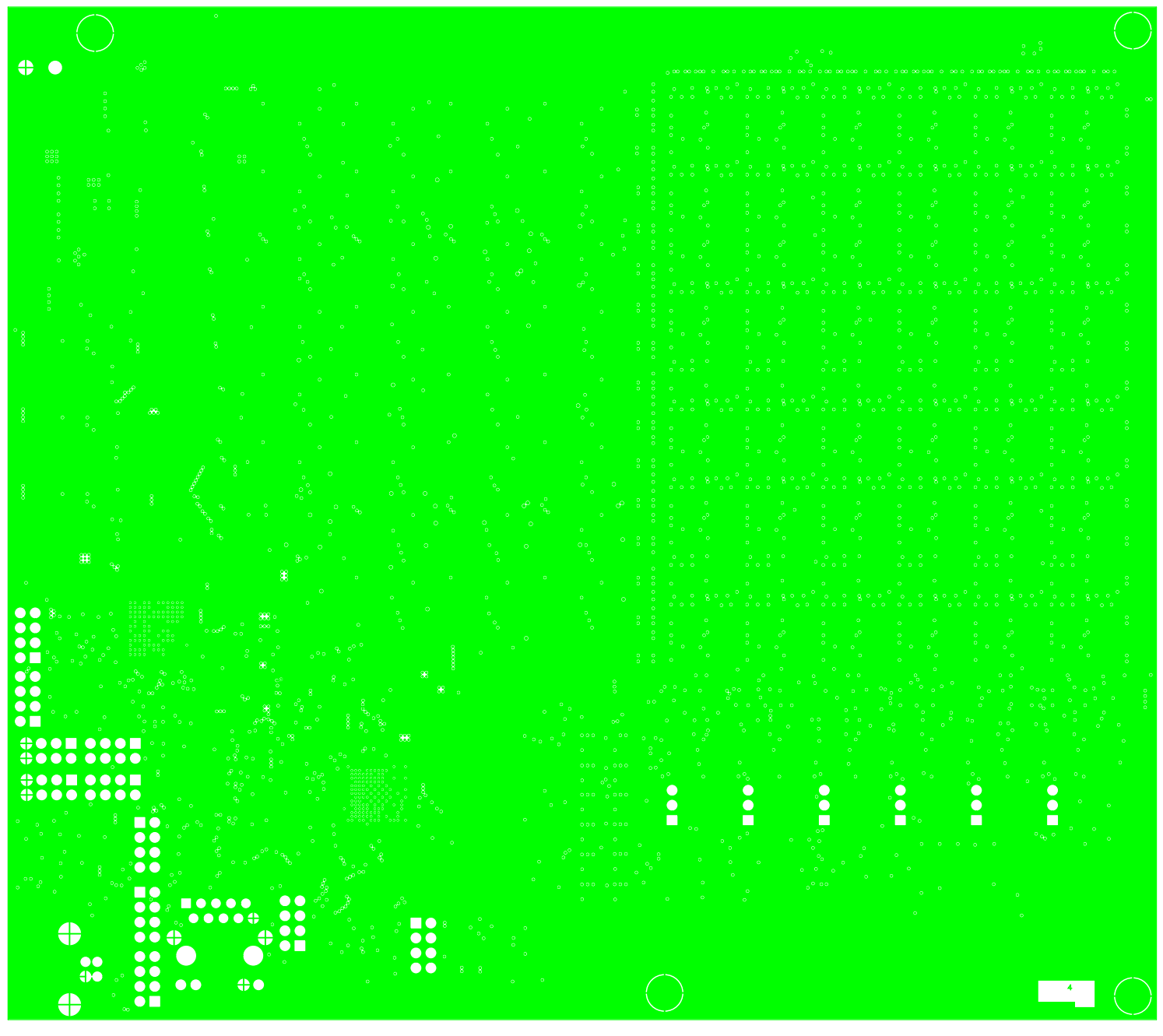
Layer Order

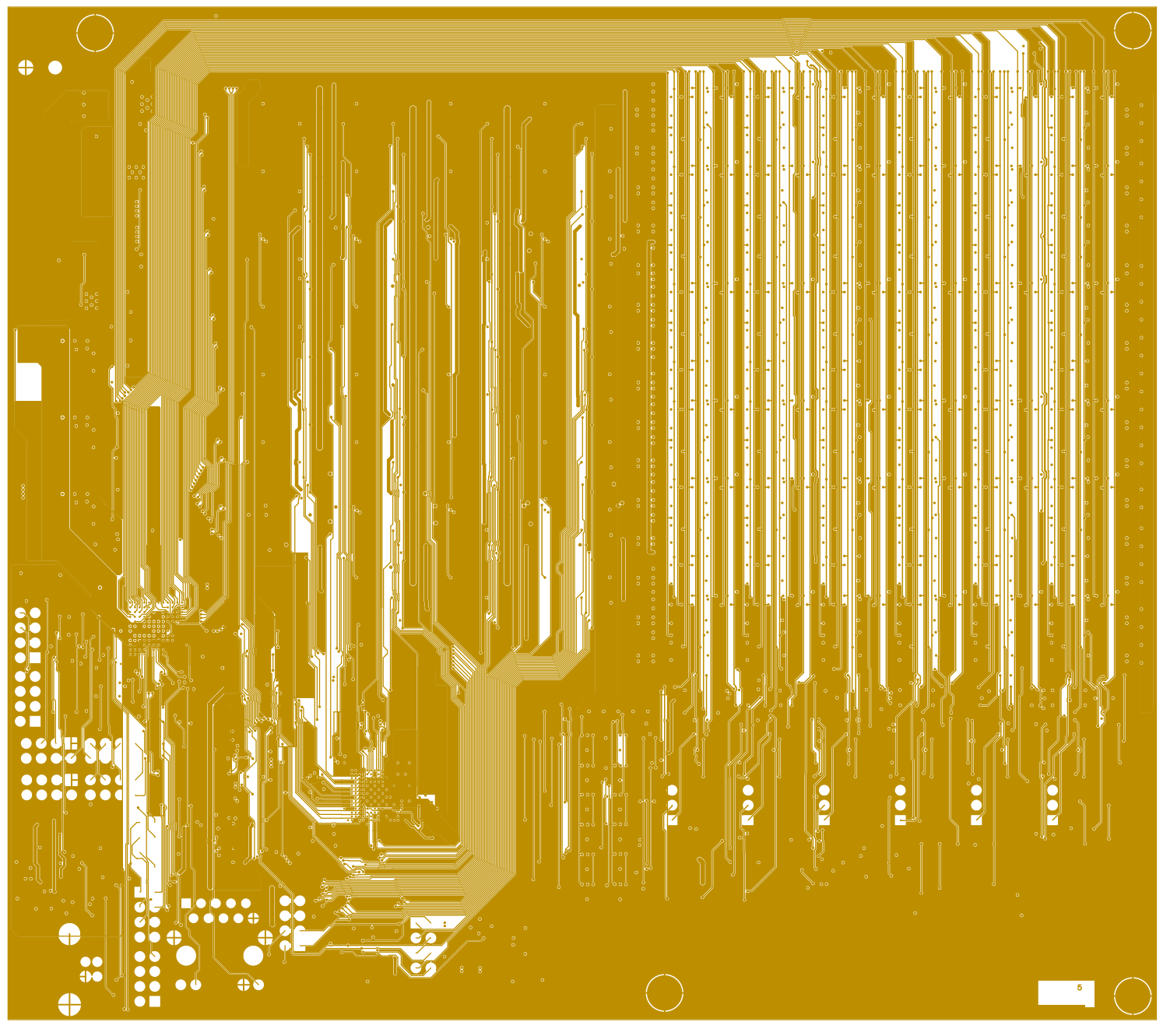


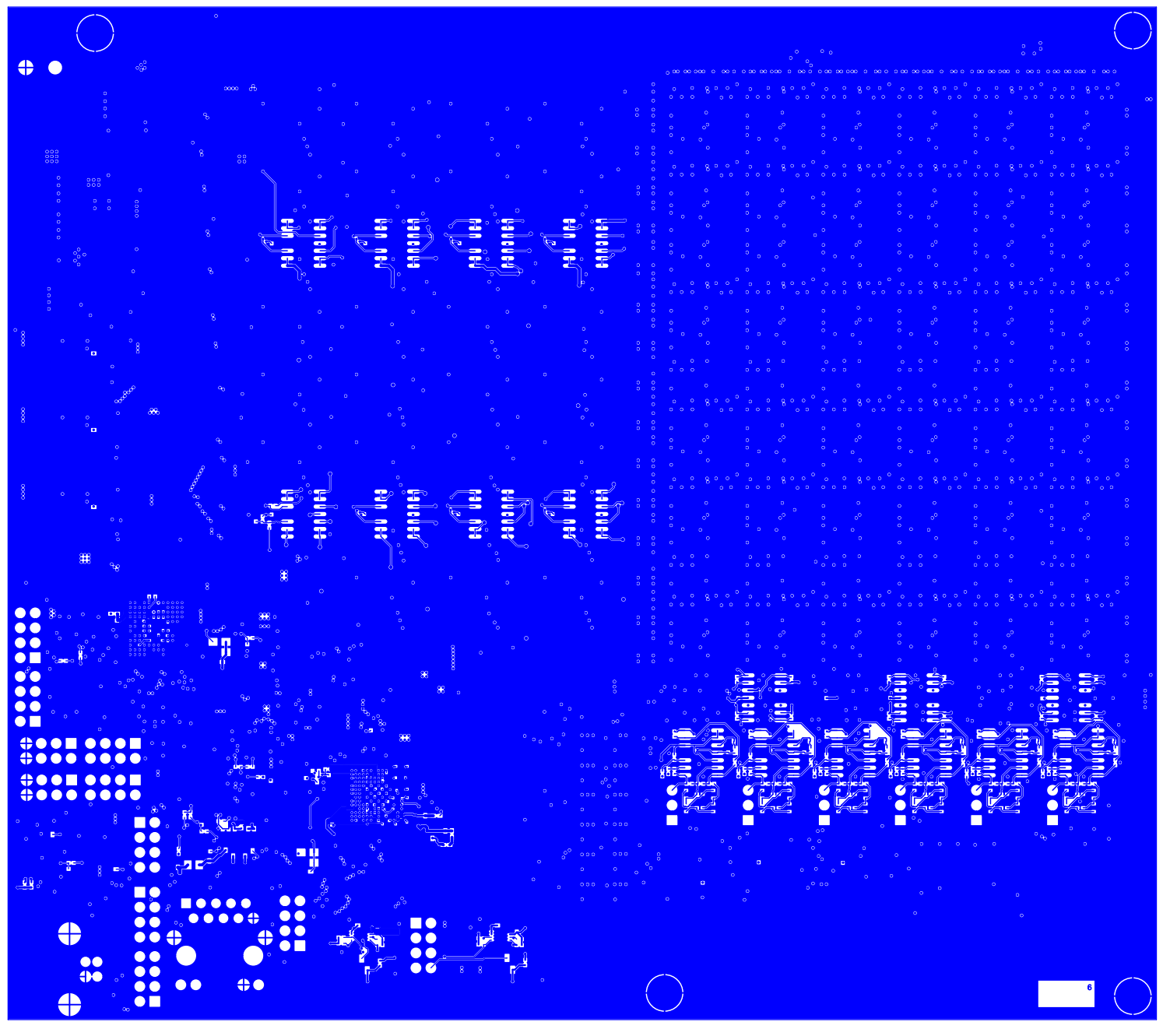


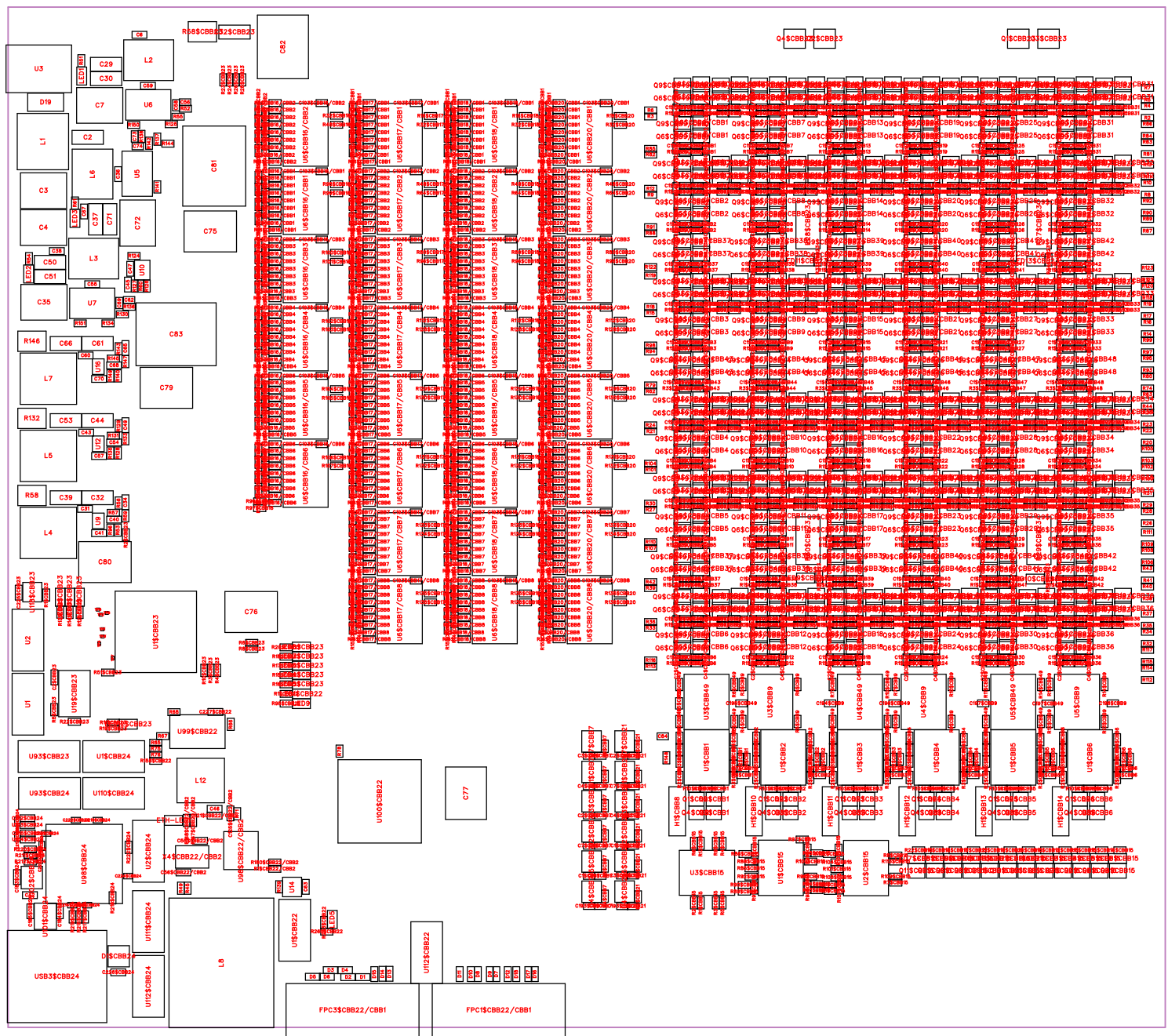


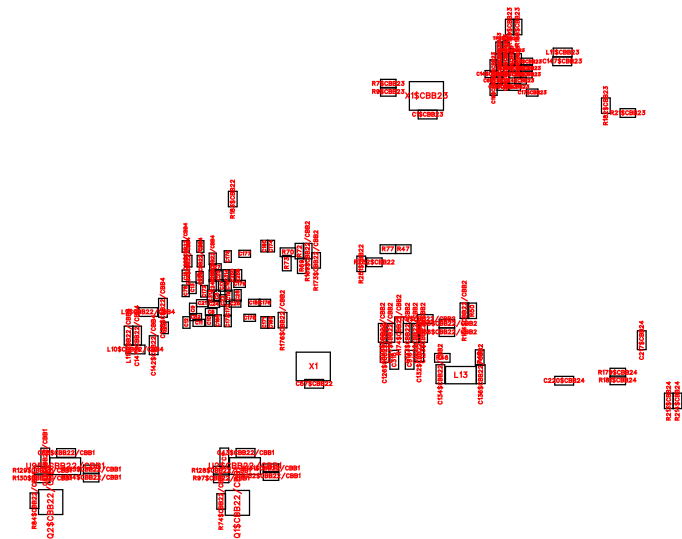
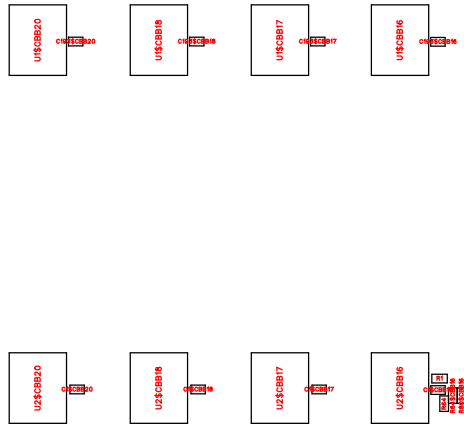
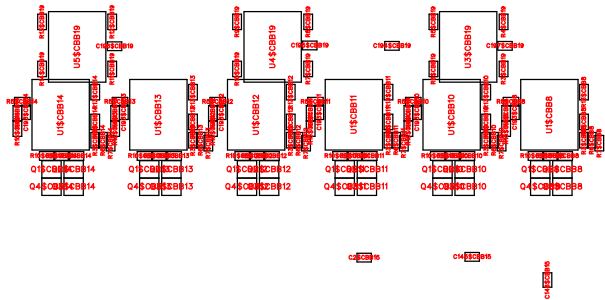












No.	Quantity	Comment	Designator	Footprint	Value	Manufacturer Part	Manufacturer	Supplier Part	Supplier	Device	Name
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1	328	100nF	C1\$CBB7,C1\$CBB9/CBB1,C1\$CBB9/CBB2,C1\$CBB9/CBB3,C1\$CBB9/CBB4,C1\$CBB9/CBB5,C1\$CBB9/CBB6,C1\$CBB9/CBB7,C1\$CBB9/CBB8,C1\$CBB9/CBB9,C1\$CBB9/CBB10,C1\$CBB9/CBB11,C1\$CBB9/CBB12,C1\$CBB9/CBB13,C1\$CBB9/CBB14,C1\$CBB9/CBB15,C1\$CBB9/CBB16,C1\$CBB9/CBB17,C1\$CBB9/CBB18,C1\$CBB9/CBB19,C1\$CBB9/CBB20,C1\$CBB9/CBB21,C1\$CBB9/CBB22,C1\$CBB9/CBB23,C1\$CBB9/CBB24,C1\$CBB9/CBB25,C1\$CBB9/CBB26,C1\$CBB9/CBB27,C1\$CBB9/CBB28,C1\$CBB9/CBB29,C1\$CBB9/CBB30,C1\$CBB9/CBB31,C1\$CBB9/CBB32,C1\$CBB9/CBB33,C1\$CBB9/CBB34,C1\$CBB9/CBB35,C1\$CBB9/CBB36,C1\$CBB9/CBB37,C1\$CBB9/CBB38,C1\$CBB9/CBB39,C1\$CBB9/CBB40,C1\$CBB9/CBB41,C1\$CBB9/CBB42,C1\$CBB16,C1\$CBB17,C1\$CBB18,C1\$CBB19/CBB1,C1\$CBB19/CBB2,C1\$CBB19/CBB3,C1\$CBB19/CBB4,C1\$CBB19/CBB5,C1\$CBB19/CBB6,C1\$CBB19/CBB7,C1\$CBB19/CBB8,C1\$CBB19/CBB9,C1\$CBB19/CBB10,C1\$CBB19/CBB11,C1\$CBB19/CBB12,C1\$CBB19/CBB13,C1\$CBB19/CBB14,C1\$CBB19/CBB15,C1\$CBB19/CBB16,C1\$CBB19/CBB17,C1\$CBB19/CBB18,C1\$CBB19/CBB19,C1\$CBB19/CBB20,C1\$CBB19/CBB21,C1\$CBB19/CBB22,C1\$CBB19/CBB23,C1\$CBB19/CBB24,C1\$CBB19/CBB25,C1\$CBB19/CBB26,C1\$CBB19/CBB27,C1\$CBB19/CBB28,C1\$CBB19/CBB29,C1\$CBB19/CBB30,C1\$CBB19/CBB31,C1\$CBB19/CBB32,C1\$CBB19/CBB33,C1\$CBB19/CBB34,C1\$CBB19/CBB35,C1\$CBB19/CBB36,C1\$CBB20,C1\$CBB21,C1\$CBB49/CBB1,C1\$CBB49/CBB2,C1\$CBB49/CBB3,C1\$CBB49/CBB4,C1\$CBB49/CBB5,C1\$CBB49/CBB6,C1\$CBB49/CBB7,C1\$CBB49/CBB8,C1\$CBB49/CBB9,C1\$CBB49/CBB10,C1\$CBB49/CBB11,C1\$CBB49/CBB12,C1\$CBB49/CBB13,C1\$CBB49/CBB14,C1\$C	C0402	100nF	CL05B104K05NNNC	SAMSUNG(三星)	C1525	LCSC	CL05B104K05NNNC	100nF
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2	26	100nF	C1,C1\$CBB23,C2\$CBB23,C36,C42,C43\$CBB22/CBB1,C46,C47,C55,C57\$CBB22,C58\$CBB22/CBB1,C59,C59\$CBB22/CBB1,C63,C123\$CBB22/CBB2,C125\$CBB22/CBB2,C126\$CBB22/CBB2,C127\$CBB22/CBB2,C128\$CBB22/CBB2,C132\$CBB22/CBB2,C133\$CBB22/CBB2,C225\$CBB23,C226\$CBB24,C227\$CBB22,C315,C316	C0402	100nF	CL05B104KB54PNC	SAMSUNG(三星)	C307331	LCSC	CL05B104KB54PNC	100nF
3	1	1uF	C2	C1206	1uF	CL31B105KBHNNNE	SAMSUNG(三星)	C1848	LCSC	CL31B105KBHNNNE	1uF

4	180	2.2nF	C2\$CBB9/CBB1,C2\$CBB9/CBB2,C2\$CBB9/CBB3,C2\$CBB9/CBB4,C2\$CBB9/CBB5,C2\$CBB9/CBB6,C2\$CBB9/CBB7,C2\$CBB9/CBB8,C2\$CBB9/CBB9,C2\$CBB9/CBB10,C2\$CBB9/CBB11,C2\$CBB9/CBB12,C2\$CBB9/CBB13,C2\$CBB9/CBB14,C2\$CBB9/CBB15,C2\$CBB9/CBB16,C2\$CBB9/CBB17,C2\$CBB9/CBB18,C2\$CBB9/CBB19,C2\$CBB9/CBB20,C2\$CBB9/CBB21,C2\$CBB9/CBB22,C2\$CBB9/CBB23,C2\$CBB9/CBB24,C2\$CBB9/CBB25,C2\$CBB9/CBB26,C2\$CBB9/CBB27,C2\$CBB9/CBB28,C2\$CBB9/CBB29,C2\$CBB9/CBB30,C2\$CBB9/CBB31,C2\$CBB9/CBB32,C2\$CBB9/CBB33,C2\$CBB9/CBB34,C2\$CBB9/CBB35,C2\$CBB9/CBB36,C2\$CBB9/CBB37,C2\$CBB9/CBB38,C2\$CBB9/CBB39,C2\$CBB9/CBB40,C2\$CBB9/CBB41,C2\$CBB9/CBB42,C2\$CBB49/CBB1,C2\$CBB49/CBB2,C2\$CBB49/CBB3,C2\$CBB49/CBB4,C2\$CBB49/CBB5,C2\$CBB49/CBB6,C2\$CBB49/CBB7,C2\$CBB49/CBB8,C2\$CBB49/CBB9,C2\$CBB49/CBB10,C2\$CBB49/CBB11,C2\$CBB49/CBB12,C2\$CBB49/CBB13,C2\$CBB49/CBB14,C2\$CBB49/CBB15,C2\$CBB49/CBB16,C2\$CBB49/CBB17,C2\$CBB49/CBB18,C2\$CBB49/CBB19,C2\$CBB49/CBB20,C2\$CBB49/CBB21,C2\$CBB49/CBB22,C2\$CBB49/CBB23,C2\$CBB49/CBB24,C2\$CBB49/CBB25,C2\$CBB49/CBB26,C2\$CBB49/CBB27,C2\$CBB49/CBB28,C2\$CBB49/CBB29,C2\$CBB49/CBB30,C2\$CBB49/CBB31,C2\$CBB49/CBB32,C2\$CBB49/CBB33,C2\$CBB49/CBB34,C2\$CBB49/CBB35,C2\$CBB49/CBB36,C2\$CBB49/CBB37,C2\$CBB49/CBB38,C2\$CBB49/CBB39,C2\$CBB49/CBB40,C2\$CBB49/CBB41,C2\$CBB49/CBB42,C2\$CBB49/CBB43,C2\$CBB49/CBB44,C2\$CBB49/CBB45,C2\$CBB49/CBB46,C2\$CBB49/CBB47,C2\$CBB49/CBB48,C4\$CBB9/CBB1,C4\$CBB9/CBB2,C4\$CBB9/CBB3,C4\$CBB9/CBB4,C4\$CBB9/CBB5,C4\$CBB9/CBB6	C0402	2.2nF	0402B222K500NT	FH(风华)	C1531	LCSC	0402B222K500NT	2.2nF
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5	78	100nF	C3\$CBB23,C4\$CBB23,C5,C5\$CBB23,C6\$CBB23,C7\$CBB23,C8,C8\$CBB23,C9,C9\$CBB23,C10,C10\$CBB23,C11,C11\$CBB23,C12,C12\$CBB23,C13,C13\$CBB23,C14,C14\$CBB23,C15,C15\$CBB23,C16,C16\$CBB23,C17,C17\$CBB23,C18,C18\$CBB23,C19,C19\$CBB23,C20,C20\$CBB23,C21,C21\$CBB23,C22,C22\$CBB22/CBB4,C22\$CBB23,C23,C23\$CBB22/CBB4,C23\$CBB23,C24,C24\$CBB22/CBB4,C25,C25\$CBB22/CBB4,C26,C27,C28,C52,C78,C168,C169,C170,C171,C172,C173,C174,C175,C176,C177,C178,C179,C180,C181,C186\$CBB24,C186\$CBB24,C213\$CBB24,C214\$CBB24,C216\$CBB24,C219\$CBB24,C221\$CBB24,C222\$CBB24,C223\$CBB24,C224\$CBB24,C248,C250,C258,C313,C314	C0201	100nF	CL03A104KP3NNNC	SAMSUNG(三星)	C49062	LCSC	CL03A104KP3NNNC	100nF
6	5	22uF	C3,C4,C7,C35,C72	C2220	22uF	FS55X226K500EHG	PSA(信昌电陶)	C784103	LCSC	FS55X226K500EHG	22uF
7	12	100pF	C5\$CBB7,C5\$CBB21,C6\$CBB7,C6\$CBB21,C8\$CBB7,C8\$CBB21,C9\$CBB7,C9\$CBB21,C10\$CBB7,C10\$CBB21,C11\$CBB7,C11\$CBB21	C0402	100pF	0402CG101J500NT	FH(风华)	C1546	LCSC	0402CG101J500NT	100pF
8	10	4.7uF	C6,C31,C38,C43,C60,C67,C212\$CBB24,C215\$CBB24,C217\$CBB24,C220\$CBB24	C0402	4.7uF	CL05A475MP5NRNC	SAMSUNG(三星)	C23733	LCSC	CL05A475MP5NRNC	4.7uF
9	13	47uF	C29,C30,C32,C32\$CBB23,C37,C39,C44,C50,C51,C53,C61,C66,C71	C1206	47uF	CL31A476M PHNNE	SAMSUNG(三星)	C96123	LCSC	CL31A476M PHNNE	47uF
10	7	4.7nF	C33,C40,C45,C48,C54,C64,C68	C0402	4.7nF	0402B472K500NT	FH(风华)	C1538	LCSC	0402B472K500NT	4.7nF
11	5	12pF	C34,C49,C55\$CBB22/CBB2,C56\$CBB22/CBB2,C65	C0402	12pF	V120J0402H QC500NBT	VIIYONG(微谷)	C5251212	LCSC	V120J0402H QC500NBT	12pF
12	3	1uF	C41,C57,C70	C0402	1uF	CL05A105KA5NQNC	SAMSUNG(三星)	C52923	LCSC	CL05A105KA5NQNC	1uF
13	3	3.3nF	C56,C62,C73	C0402	3.3nF	CL05B332KB5NNNC	SAMSUNG(三星)	C26404	LCSC	CL05B332KB5NNNC	3.3nF
14	3	47pF	C58,C69,C74	C0402	47pF	0402CG470J500NT	FH(风华)	C1567	LCSC	0402CG470J500NT	47pF
15	5	330uF	C75,C76,C77,C79,C80	CAP-SMD_B D6.3-L6.6-W6.6-FD	330uF	RV70J331M0607	HONOR(荣誉)	C58276	LCSC	RV70J331M0607_C58276	330uF

16	2	100uF	C81,C83	CAP-SMD_B D10.0-L10.3- W10.3-LS11. 0-FD	100uF	MVK 50V100 M 10*10	SAMYOUNG (韩国三莹)	C165627	LCSC	MVK 50V100 M 10*10	100uF
17	1	330uF	C82	CAP-SMD_B D8.0-L8.3-W 8.3-FD	330uF	RVT1E331M 0810 330UF 25V	DMBJ(振宝佳)	C970701	LCSC	RVT1E331M 0810 330UF 25V	330uF
18	4	4.7uF	C121\$CBB22/CBB2,C122\$ CBB22/CBB2,C134\$CBB22 /CBB2,C136\$CBB22/CBB2	C0402	4.7uF	LMK105BBJ 475MVLf	TAIYO YUDE N(太诱)	C23733	LCSC	LMK105BBJ 475MVLf	4.7uF
19	4	10uF	C138\$CBB22/CBB4,C142\$ CBB22/CBB4,C147\$CBB22 /CBB4,C147\$CBB23	C0402	10uF	CL05A106M Q5NUNC	SAMSUNG(三星)	C15525	LCSC	CL05A106M Q5NUNC	10uF
20	2	10pF	C184\$CBB24,C185\$CBB24	C0402	10pF	CL05C100JB 5NNNC	SAMSUNG(三星)	C32949	LCSC	CL05C100JB 5NNNC	10pF
21	1	USBLC6-2S C6	D1\$CBB24	SOT-23-6_L2 .9-W1.6-P0.9 5-LS2.8-BL		USBLC6-2S C6	ST(意法半导 体)	C7519	LCSC	USBLC6-2S C6_C7519	USBLC6-2S C6
22	1	RBR5L60AD DTE25	D19	SMA_L4.5-W 2.6-L55.0-R D		RBR5L60AD DTE25	ROHM(罗姆)	C5343112	LCSC	RBR5L60AD DTE25	RBR5L60AD DTE25
23	32	KT-0603R	ETH-LED1,LED1,LED1\$CB B15,LED1\$CBB22,LED1\$C BB23,LED2,LED2\$CBB15,L ED2\$CBB23,LED3,LED3\$C BB15,LED3\$CBB23,LED4\$ CBB15,LED4\$CBB23,LED5 ,LED5\$CBB15,LED5\$CBB2 3,LED6\$CBB15,LED6\$CBB 23,LED7\$CBB15,LED7\$CB B23,LED8\$CBB15,LED8\$C BB23,LED9,LED9\$CBB15,L ED9\$CBB23,LED10\$CBB1 5,LED10\$CBB23,LED11\$C BB15,LED11\$CBB23,LED1 2\$CBB15,LED12\$CBB23,L ED13\$CBB23	LED0603-RD		KT-0603R	KENTO	C2286	LCSC	KT-0603R	KT-0603R
24	2	522711579	FPC1\$CBB22/CBB1,FPC3\$ CBB22/CBB1	FPC-SMD_5 22711579		522711579	MOLEX	C7545783	LCSC	522711579	522711579
25	6	PZ254-1-03- Z-8.5	H1\$CBB8,H1\$CBB10,H1\$C BB11,H1\$CBB12,H1\$CBB1 3,H1\$CBB14	HDR-TH_03 P-P2.54-V-M		PZ254-1-03- Z-8.5	HCTL(华灿天 禄)	C2894926	LCSC	PZ254-1-03- Z-8.5_C2894 926	PZ254-1-03- Z-8.5
26	4	1.5uH	L1,L4,L5,L7	IND-SMD_L7 .8-W7.0_GC D75	1.5uH	CD75-1R5M 1.5UH	DMBJ(振宝佳)	C2826688	LCSC	CD75-1R5M 1.5UH	1.5uH
27	3	6.8uH	L2,L3,L6	IND-SMD_L6 .6-W6.6	6.8uH	SMMS0630- 6R8M	SXN(顺翔诺)	C128690	LCSC	SMMS0630- 6R8M	6.8uH
28	1	HR911130C	L8	RJ45-TH_HR 911130C		HR911130C	HANRUN(汉 仁)	C50933	LCSC	HR911130C	HR911130C
29	4	BLM15AX22 1SN1D	L9\$CBB22/CBB4,L10\$CBB 22/CBB4,L11\$CBB22/CBB4 ,L11\$CBB23	L0402		BLM15AX22 1SN1D	muRata(村田)	C76885	LCSC	BLM15AX22 1SN1D	BLM15AX22 1SN1D

30	1	4.7uH	L12	IND-SMD L7.8-W7.0 SU MIDA_CD73	4.7uH	FCD75-4R7 M	cjiang(长江微电)	C41382395	LCSC	FCD75-4R7 M	4.7uH
31	1	APBD2012-152WTD12	L13	L0805		APBD2012-152WTD12	APV(爱普微)	C50325953	LCSC	APBD2012-152WTD12	APBD2012-152WTD12
32	48	MMBT3904 WT1G	Q1\$CBB1,Q1\$CBB2,Q1\$CBB3,Q1\$CBB4,Q1\$CBB5,Q1\$CBB6,Q1\$CBB8,Q1\$CBB10,Q1\$CBB11,Q1\$CBB12,Q1\$CBB13,Q1\$CBB14,Q2\$CBB1,Q2\$CBB2,Q2\$CBB3,Q2\$CBB4,Q2\$CBB5,Q2\$CBB6,Q2\$CBB8,Q2\$CBB10,Q2\$CBB11,Q2\$CBB12,Q2\$CBB13,Q2\$CBB14,Q3\$CBB1,Q3\$CBB2,Q3\$CBB3,Q3\$CBB4,Q3\$CBB5,Q3\$CBB6,Q3\$CBB8,Q3\$CBB10,Q3\$CBB11,Q3\$CBB12,Q3\$CBB13,Q3\$CBB14,Q4\$CBB1,Q4\$CBB2,Q4\$CBB3,Q4\$CBB4,Q4\$CBB5,Q4\$CBB6,Q4\$CBB8,Q4\$CBB10,Q4\$CBB11,Q4\$CBB12,Q4\$CBB13,Q4\$CBB14	SC-70-3 L2.1-W1.3-P1.3 0-LS2.1-BR		MMBT3904 WT1G	onsemi(安森美)	C58011	LCSC	MMBT3904 WT1G	MMBT3904 WT1G

33	516	DOX3134A	Q1\$CBB15,Q2\$CBB15,Q3\$CBB15,Q4\$CBB15,Q5\$CBB9/CBB1,Q5\$CBB9/CBB2,Q5\$CBB9/CBB3,Q5\$CBB9/CBB4,Q5\$CBB9/CBB5,Q5\$CBB9/CBB6,Q5\$CBB9/CBB7,Q5\$CBB9/CBB8,Q5\$CBB9/CBB9,Q5\$CBB9/CBB10,Q5\$CBB9/CBB11,Q5\$CBB9/CBB12,Q5\$CBB9/CBB13,Q5\$CBB9/CBB14,Q5\$CBB9/CBB15,Q5\$CBB9/CBB16,Q5\$CBB9/CBB17,Q5\$CBB9/CBB18,Q5\$CBB9/CBB19,Q5\$CBB9/CBB20,Q5\$CBB9/CBB21,Q5\$CBB9/CBB22,Q5\$CBB9/CBB23,Q5\$CBB9/CBB24,Q5\$CBB9/CBB25,Q5\$CBB9/CBB26,Q5\$CBB9/CBB27,Q5\$CBB9/CBB28,Q5\$CBB9/CBB29,Q5\$CBB9/CBB30,Q5\$CBB9/CBB31,Q5\$CBB9/CBB32,Q5\$CBB9/CBB33,Q5\$CBB9/CBB34,Q5\$CBB9/CBB35,Q5\$CBB9/CBB36,Q5\$CBB9/CBB37,Q5\$CBB9/CBB38,Q5\$CBB9/CBB39,Q5\$CBB9/CBB40,Q5\$CBB9/CBB41,Q5\$CBB9/CBB42,Q5\$CBB15,Q5\$CBB19/CBB1,Q5\$CBB19/CBB2,Q5\$CBB19/CBB3,Q5\$CBB19/CBB4,Q5\$CBB19/CBB5,Q5\$CBB19/CBB6,Q5\$CBB19/CBB7,Q5\$CBB19/CBB8,Q5\$CBB19/CBB9,Q5\$CBB19/CBB10,Q5\$CBB19/CBB11,Q5\$CBB19/CBB12,Q5\$CBB19/CBB13,Q5\$CBB19/CBB14,Q5\$CBB19/CBB15,Q5\$CBB19/CBB16,Q5\$CBB19/CBB17,Q5\$CBB19/CBB18,Q5\$CBB19/CBB19,Q5\$CBB19/CBB20,Q5\$CBB19/CBB21,Q5\$CBB19/CBB22,Q5\$CBB19/CBB23,Q5\$CBB19/CBB24,Q5\$CBB19/CBB25,Q5\$CBB19/CBB26,Q5\$CBB19/CBB27,Q5\$CBB19/CBB28,Q5\$CBB19/CBB29,Q5\$CBB19/CBB30,Q5\$CBB19/CBB31,Q5\$CBB19/CBB32,Q5\$CBB19/CBB33,Q5\$CBB19/CBB34,Q5\$CBB19/CBB35,Q5\$CBB49/CBB1,Q5\$CBB49/CBB2,Q5\$CBB49/CBB3,Q5\$CBB49/CBB4,Q5\$CBB49/CBB5,Q5\$CBB49/CBB6,Q5\$CBB49/CBB7,Q5\$CBB49/CBB8,Q5\$CBB49/CBB9,Q5\$CBB49/CBB10,Q5\$CBB49/CBB11,Q5\$CBB49/CBB12,Q5\$CBB49/CBB13,Q5\$CBB49/CBB14,Q5\$CBB4	SOT-323-3 L 2.1-W1.3-P1. 30-LS2.3-BR		DOX3134A	DOINGTER(杜因特)	C534349756	LCSC	DOX3134A	DOX3134A
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34	2	BSS138	Q1\$CBB22/CBB1,Q2\$CBB22/CBB1	SOT-23-3 L2 .9-W1.3-P1.9 0-LS2.4-BR-1		BSS138	KUU	C3015220	LCSC	BSS138	BSS138
35	4	DO2320AA	Q1\$CBB23,Q2\$CBB23,Q3\$CBB23,Q4\$CBB23	SOT-23-3 L2 .9-W1.6-P1.9 0-LS2.8-BR		DO2320AA	DOINGTER(杜因特)	C47018574	LCSC	DO2320AA	DO2320AA
36	6	300Ω	R1\$CBB1,R1\$CBB2,R1\$CBB3,R1\$CBB4,R1\$CBB5,R1\$CBB6	R0402	300Ω	0402WGF300TCE	UNI-ROYAL(厚声)	C25102	LCSC	0402WGF300TCE	300Ω
37	60	100Ω	R1\$CBB7,R1\$CBB21,R2\$CBB7,R2\$CBB21,R3\$CBB7,R3\$CBB9,R3\$CBB19,R3\$CBB21,R3\$CBB49,R4\$CBB7,R4\$CBB9,R4\$CBB19,R4\$CBB21,R4\$CBB49,R5\$CBB7,R5\$CBB9,R5\$CBB19,R5\$CBB21,R5\$CBB49,R6\$CBB7,R6\$CBB9,R6\$CBB19,R6\$CBB21,R6\$CBB49,R7\$CBB9,R7\$CBB19,R7\$CBB49,R8\$CBB9,R8\$CBB19,R8\$CBB49,R9\$CBB9,R9\$CBB19,R9\$CBB49,R10\$CBB1,R10\$CBB2,R10\$CBB3,R10\$CBB4,R10\$CBB5,R10\$CBB6,R10\$CBB8,R10\$CBB9,R10\$CBB10,R10\$CBB11,R10\$CBB12,R10\$CBB13,R10\$CBB14,R10\$CBB19,R10\$CBB49,R11\$CBB9,R11\$CBB19,R11\$CBB49,R12\$CBB9,R12\$CBB19,R12\$CBB49,R13\$CBB9,R13\$CBB19,R13\$CBB49,R14\$CBB9,R14\$CBB49	R0402	100Ω	0402WGF100TCE	UNI-ROYAL(厚声)	C25076	LCSC	0402WGF100TCE	100Ω

38	240	1MΩ	R1\$CBB9/CBB1,R1\$CBB9/CBB2,R1\$CBB9/CBB3,R1\$CBB9/CBB4,R1\$CBB9/CBB5,R1\$CBB9/CBB6,R1\$CBB9/CBB7,R1\$CBB9/CBB8,R1\$CBB9/CBB9,R1\$CBB9/CBB10,R1\$CBB9/CBB11,R1\$CBB9/CBB12,R1\$CBB9/CBB13,R1\$CBB9/CBB14,R1\$CBB9/CBB15,R1\$CBB9/CBB16,R1\$CBB9/CBB17,R1\$CBB9/CBB18,R1\$CBB9/CBB19,R1\$CBB9/CBB20,R1\$CBB9/CBB21,R1\$CBB9/CBB22,R1\$CBB9/CBB23,R1\$CBB9/CBB24,R1\$CBB9/CBB25,R1\$CBB9/CBB26,R1\$CBB9/CBB27,R1\$CBB9/CBB28,R1\$CBB9/CBB29,R1\$CBB9/CBB30,R1\$CBB9/CBB31,R1\$CBB9/CBB32,R1\$CBB9/CBB33,R1\$CBB9/CBB34,R1\$CBB9/CBB35,R1\$CBB9/CBB36,R1\$CBB9/CBB37,R1\$CBB9/CBB38,R1\$CBB9/CBB39,R1\$CBB9/CBB40,R1\$CBB9/CBB41,R1\$CBB9/CBB42,R1\$CBB19/CBB1,R1\$CBB19/CBB2,R1\$CBB19/CBB3,R1\$CBB19/CBB4,R1\$CBB19/CBB5,R1\$CBB19/CBB6,R1\$CBB19/CBB7,R1\$CBB19/CBB8,R1\$CBB19/CBB9,R1\$CBB19/CBB10,R1\$CBB19/CBB11,R1\$CBB19/CBB12,R1\$CBB19/CBB13,R1\$CBB19/CBB14,R1\$CBB19/CBB15,R1\$CBB19/CBB16,R1\$CBB19/CBB17,R1\$CBB19/CBB18,R1\$CBB19/CBB19,R1\$CBB19/CBB20,R1\$CBB19/CBB21,R1\$CBB19/CBB22,R1\$CBB19/CBB23,R1\$CBB19/CBB24,R1\$CBB19/CBB25,R1\$CBB19/CBB26,R1\$CBB19/CBB27,R1\$CBB19/CBB28,R1\$CBB19/CBB29,R1\$CBB19/CBB30,R1\$CBB19/CBB31,R1\$CBB19/CBB32,R1\$CBB19/CBB33,R1\$CBB19/CBB34,R1\$CBB19/CBB35,R1\$CBB19/CBB36,R1\$CBB19/CBB37,R1\$CBB19/CBB38,R1\$CBB19/CBB39,R1\$CBB19/CBB40,R1\$CBB19/CBB41,R1\$CBB19/CBB42,R1\$CBB49/CBB1,R1\$CBB49/CBB2,R1\$CBB49/CBB3,R1\$CBB49/CBB4,R1\$CBB49/CBB5,R1\$CBB49/CBB6,R1\$CBB49/CBB7,R1\$CBB49/CBB8,R1\$CBB49/CBB9,R1\$CBB49/CBB10,R1\$CBB49/CBB11,R1\$CBB49/CBB12,R1\$CBB49/CBB13,R1\$CBB49/CBB14,R1\$CBB49/CBB15,R1\$CBB49/CBB16,R1\$CBB49/CBB17,R1\$CBB49/C	R0402	1MΩ	0402WGF10 04TCE	UNI-ROYAL(厚声)	C26083	LCSC	0402WGF10 04TCE	1MΩ
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39	91	1kΩ	R1\$CBB15,R4\$CBB1,R4\$CBB2,R4\$CBB3,R4\$CBB4,R4\$CBB5,R4\$CBB6,R4\$CBB8,R4\$CBB10,R4\$CBB11,R4\$CBB12,R4\$CBB13,R4\$CBB14,R5\$CBB1,R5\$CBB2,R5\$CBB3,R5\$CBB4,R5\$CBB5,R5\$CBB6,R5\$CBB8,R5\$CBB10,R5\$CBB11,R5\$CBB12,R5\$CBB13,R5\$CBB14,R5\$CBB15,R6\$CBB1,R6\$CBB2,R6\$CBB3,R6\$CBB4,R6\$CBB5,R6\$CBB6,R6\$CBB8,R6\$CBB10,R6\$CBB11,R6\$CBB12,R6\$CBB13,R6\$CBB14,R7\$CBB1,R7\$CBB2,R7\$CBB3,R7\$CBB4,R7\$CBB5,R7\$CBB6,R7\$CBB8,R7\$CBB10,R7\$CBB11,R7\$CBB12,R7\$CBB13,R7\$CBB14,R9\$CBB1,R9\$CBB2,R9\$CBB3,R9\$CBB4,R9\$CBB5,R9\$CBB6,R9\$CBB8,R9\$CBB10,R9\$CBB11,R9\$CBB12,R9\$CBB13,R9\$CBB14,R9\$CBB15,R13\$CBB8,R13\$CBB10,R13\$CBB11,R13\$CBB12,R13\$CBB13,R13\$CBB14,R13\$CBB15,R15\$CBB8,R15\$CBB10,R15\$CBB11,R15\$CBB12,R15\$CBB13,R15\$CBB14,R85\$CBB15,R86\$CBB15,R89\$CBB15,R90\$CBB15,R93\$CBB15,R94\$CBB15,R97\$CBB15,R98\$CBB15,R101\$CBB15,R102\$CBB15,R105\$CBB15,R106\$CBB15,R109\$CBB15,R113\$CBB15,R218\$CBB24	R0402	1kΩ	0402WGF10 01TCE	UNI-ROYAL(厚声)	C11702	LCSC	0402WGF10 01TCE	1kΩ
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40	270	20kΩ	R1\$CBB16/CBB1,R1\$CBB16/CBB2,R1\$CBB16/CBB3,R1\$CBB16/CBB4,R1\$CBB16/CBB5,R1\$CBB16/CBB6,R1\$CBB17/CBB1,R1\$CBB17/CBB2,R1\$CBB17/CBB3,R1\$CBB17/CBB4,R1\$CBB17/CBB5,R1\$CBB17/CBB6,R1\$CBB17/CBB7,R1\$CBB17/CBB8,R1\$CBB18/CBB1,R1\$CBB18/CBB2,R1\$CBB18/CBB3,R1\$CBB18/CBB4,R1\$CBB18/CBB5,R1\$CBB18/CBB6,R1\$CBB18/CBB7,R1\$CBB18/CBB8,R1\$CBB20/CBB1,R1\$CBB20/CBB2,R1\$CBB20/CBB3,R1\$CBB20/CBB4,R1\$CBB20/CBB5,R1\$CBB20/CBB6,R1\$CBB20/CBB7,R1\$CBB20/CBB8,R3\$CBB16/CBB1,R3\$CBB16/CBB2,R3\$CBB16/CBB3,R3\$CBB16/CBB4,R3\$CBB16/CBB5,R3\$CBB16/CBB6,R3\$CBB17/CBB1,R3\$CBB17/CBB2,R3\$CBB17/CBB3,R3\$CBB17/CBB4,R3\$CBB17/CBB5,R3\$CBB17/CBB6,R3\$CBB17/CBB7,R3\$CBB17/CBB8,R3\$CBB18/CBB1,R3\$CBB18/CBB2,R3\$CBB18/CBB4,R3\$CBB18/CBB5,R3\$CBB18/CBB6,R3\$CBB18/CBB7,R3\$CBB18/CBB8,R3\$CBB20/CBB1,R3\$CBB20/CBB2,R3\$CBB20/CBB3,R3\$CBB20/CBB4,R3\$CBB20/CBB5,R3\$CBB20/CBB6,R3\$CBB20/CBB7,R3\$CBB20/CBB8,R4\$CBB16/CBB1,R4\$CBB16/CBB2,R4\$CBB16/CBB3,R4\$CBB16/CBB4,R4\$CBB16/CBB5,R4\$CBB16/CBB6,R4\$CBB17/CBB1,R4\$CBB17/CBB2,R4\$CBB17/CBB3,R4\$CBB17/CBB4,R4\$CBB17/CBB5,R4\$CBB17/CBB6,R4\$CBB17/CBB7,R4\$CBB17/CBB8,R4\$CBB18/CBB1,R4\$CBB18/CBB2,R4\$CBB18/CBB4,R4\$CBB18/CBB5,R4\$CBB18/CBB6,R4\$CBB18/CBB7,R4\$CBB18/CBB8,R4\$CBB20/CBB1,R4\$CBB20/CBB2,R4\$CBB20/CBB3,R4\$CBB20/CBB4,R4\$CBB20/CBB5,R4\$CBB20/CBB6,R4\$CBB20/CBB7,R4\$CBB20/CBB8,R5\$CBB16/CBB1,R5\$CBB16/CBB3,R5\$CBB16/CBB4,R5\$CBB16/CBB5,R5\$CBB16/CBB6,R5\$CBB17/CBB1,R5\$CBB17/	R0402	20kΩ	ARG02BTC2002	Viking(光碩)	C2984413	LCSC	ARG02BTC2002	20kΩ
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41	34	470Ω	R1\$CBB22,R10\$CBB15,R11\$CBB15,R12\$CBB15,R12\$CBB23,R13\$CBB23,R14\$CBB15,R14\$CBB23,R15\$CBB15,R15\$CBB23,R16\$CBB15,R16\$CBB23,R17\$CBB15,R17\$CBB23,R18\$CBB15,R18\$CBB23,R19\$CBB15,R19\$CBB23,R20\$CBB15,R20\$CBB23,R21\$CBB15,R22\$CBB15,R31\$CBB23,R32\$CBB23,R33\$CBB23,R34\$CBB23,R49,R51,R54,R61,R95\$CBB22,R166\$CBB22/CBB2,R168\$CBB22/CBB2,R260\$CBB22	R0402	470Ω	0402WGF4700TCE	UNI-ROYAL(厚声)	C25117	LCSC	0402WGF4700TCE	470Ω
42	39	4.7kΩ	R1\$CBB22/CBB2,R16\$CBB17,R16\$CBB18,R16\$CBB20,R48\$CBB17,R48\$CBB18,R48\$CBB20,R50,R74\$CBB22/CBB1,R80\$CBB17,R80\$CBB18,R80\$CBB20,R84\$CBB22/CBB1,R97\$CBB22/CBB1,R112\$CBB17,R112\$CBB18,R112\$CBB20,R128\$CBB22/CBB1,R129\$CBB17,R129\$CBB18,R129\$CBB20,R129\$CBB22/CBB1,R130\$CBB22/CBB1,R131\$CBB17,R131\$CBB18,R131\$CBB20,R133\$CBB17,R133\$CBB18,R133\$CBB20,R135\$CBB17,R135\$CBB20,R137,R165\$CBB22/CBB2,R167\$CBB22/CBB2,R169\$CBB22/CBB2,R179\$CBB22/CBB2,R179\$CBB24,R180\$CBB22/CBB2	R0402	4.7kΩ	0402WGF4701TCE	UNI-ROYAL(厚声)	C25900	LCSC	0402WGF4701TCE	4.7kΩ
43	28	10kΩ	R1\$CBB23,R2\$CBB23,R3\$CBB23,R4\$CBB23,R5\$CBB23,R6\$CBB23,R7\$CBB23,R11\$CBB23,R23\$CBB23,R24\$CBB23,R25\$CBB23,R26\$CBB23,R66,R67,R68,R69,R70,R71,R124,R139,R181\$CBB24,R211\$CBB24,R213\$CBB24,R214\$CBB24,R220\$CBB24,R222\$CBB24,R252\$CBB22,R253\$CBB22	R0402	10kΩ	0402WGF1002TCE	UNI-ROYAL(厚声)	C25744	LCSC	0402WGF1002TCE	10kΩ
44	12	330kΩ	R1\$CBB49/CBB43,R1\$CBB49/CBB44,R1\$CBB49/CBB45,R1\$CBB49/CBB46,R1\$CBB49/CBB47,R1\$CBB49/CBB48,R3\$CBB49/CBB43,R3\$CBB49/CBB44,R3\$CBB49/CBB45,R3\$CBB49/CBB46,R3\$CBB49/CBB47,R3\$CBB49/CBB48	R0402	330kΩ	RTT02334JTH	RALEC(旺途)	C11887	LCSC	RTT02334JTH	330kΩ

45	4	20kΩ	R1,R64,R64\$CBB16,R96\$CBB16	R0402	20kΩ	0402WGF2002TCE	UNI-ROYAL(厚声)	C25765	LCSC	0402WGF2002TCE	20kΩ
46	12	1.2kΩ	R2\$CBB1,R2\$CBB2,R2\$CBB3,R2\$CBB4,R2\$CBB5,R2\$CBB6,R2\$CBB8,R2\$CBB10,R2\$CBB11,R2\$CBB12,R2\$CBB13,R2\$CBB14	R0402	1.2kΩ	0402WGF1201TCE	UNI-ROYAL(厚声)	C25862	LCSC	0402WGF1201TCE	1.2kΩ
47	6	3.9kΩ	R2\$CBB15,R3\$CBB15,R4\$CBB15,R6\$CBB15,R7\$CBB15,R8\$CBB15	R0402	3.9kΩ	0402WGF3901TCE	UNI-ROYAL(厚声)	C51721	LCSC	0402WGF3901TCE	3.9kΩ

48	210	10kΩ	R2\$CBB16/CBB1,R2\$CBB16/CBB2,R2\$CBB16/CBB3,R2\$CBB16/CBB4,R2\$CBB16/CBB5,R2\$CBB16/CBB6,R2\$CBB17/CBB1,R2\$CBB17/CBB2,R2\$CBB17/CBB3,R2\$CBB17/CBB4,R2\$CBB17/CBB5,R2\$CBB17/CBB6,R2\$CBB17/CBB7,R2\$CBB17/CBB8,R2\$CBB18/CBB1,R2\$CBB18/CBB2,R2\$CBB18/CBB3,R2\$CBB18/CBB4,R2\$CBB18/CBB5,R2\$CBB18/CBB6,R2\$CBB18/CBB7,R2\$CBB18/CBB8,R2\$CBB20/CBB1,R2\$CBB20/CBB2,R2\$CBB20/CBB3,R2\$CBB20/CBB4,R2\$CBB20/CBB5,R2\$CBB20/CBB6,R2\$CBB20/CBB7,R2\$CBB20/CBB8,R10\$CBB16/CBB1,R10\$CBB16/CBB2,R10\$CBB16/CBB3,R10\$CBB16/CBB4,R10\$CBB16/CBB5,R10\$CBB16/CBB6,R10\$CBB17/CBB1,R10\$CBB17/CBB2,R10\$CBB17/CBB3,R10\$CBB17/CBB4,R10\$CBB17/CBB5,R10\$CBB17/CBB6,R10\$CBB17/CBB7,R10\$CBB17/CBB8,R11\$CBB16/CBB1,R11\$CBB16/CBB2,R11\$CBB16/CBB3,R11\$CBB16/CBB4,R11\$CBB16/CBB5,R11\$CBB16/CBB6,R11\$CBB17/CBB1,R11\$CBB17/CBB2,R11\$CBB17/CBB3,R11\$CBB17/CBB4,R11\$CBB17/CBB5,R11\$CBB17/CBB6,R11\$CBB17/CBB7,R11\$CBB17/CBB8,R11\$CBB18/CBB1,R11\$CBB18/CBB2,R11\$CBB18/CBB3,R11\$CBB18/CBB4,R11\$CBB18/CBB5,R11\$CBB18/CBB6,R11\$CBB18/CBB7,R11\$CBB18/CBB8,R12\$CBB16/CBB1,R12\$CBB16/CBB2,R12\$CBB16/CBB3,R12\$	R0402	10kΩ	RT0402BRD0710KL	YAGEO(国巨)	C190095	LCSC	RT0402BRD0710KL	10kΩ
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49	60	120Ω	R2,R4,R5,R7,R8,R10,R11,R13,R14,R16,R17,R19,R20,R22,R23,R25,R26,R28,R29,R31,R32,R34,R35,R37,R38,R40,R41,R43,R44,R63,R74,R80,R81,R83,R84,R86,R87,R89,R90,R92,R93,R95,R97,R99,R100,R102,R103,R105,R106,R108,R109,R111,R112,R114,R115,R117,R118,R120,R121,R123	R0402	120Ω	0402WGF1200TCE	UNI-ROYAL(厚声)	C25079	LCSC	0402WGF1200TCE	120Ω
50	24	910Ω	R3\$CBB1,R3\$CBB2,R3\$CBB3,R3\$CBB4,R3\$CBB5,R3\$CBB6,R3\$CBB8,R3\$CBB10,R3\$CBB11,R3\$CBB12,R3\$CBB13,R3\$CBB14,R8\$CBB1,R8\$CBB2,R8\$CBB3,R8\$CBB4,R8\$CBB5,R8\$CBB6,R8\$CBB8,R8\$CBB10,R8\$CBB11,R8\$CBB12,R8\$CBB13,R8\$CBB14	R0402	910Ω	SCR0402J910R	VO(翔胜)	C3017583	LCSC	SCR0402J910R	910Ω
51	30	68Ω	R3,R6,R9,R12,R15,R18,R21,R24,R27,R30,R33,R36,R39,R42,R62,R79,R82,R85,R88,R91,R94,R98,R101,R104,R107,R110,R113,R116,R119,R122	R0402	68Ω	0402WGJ0680TCE	UNI-ROYAL(厚声)	C25178	LCSC	0402WGJ0680TCE	68Ω
52	30	3.3kΩ	R11\$CBB1,R11\$CBB2,R11\$CBB3,R11\$CBB4,R11\$CBB5,R11\$CBB6,R12\$CBB1,R12\$CBB2,R12\$CBB3,R12\$CBB4,R12\$CBB5,R12\$CBB6,R13\$CBB1,R13\$CBB2,R13\$CBB3,R13\$CBB4,R13\$CBB5,R13\$CBB6,R67\$CBB15,R68\$CBB15,R91\$CBB15,R92\$CBB15,R95\$CBB15,R96\$CBB15,R99\$CBB15,R100\$CBB15,R103\$CBB15,R104\$CBB15,R107\$CBB15,R108\$CBB15	R0402	3.3kΩ	0402WGF3301TCE	UNI-ROYAL(厚声)	C25890	LCSC	0402WGF3301TCE	3.3kΩ
53	6	270Ω	R14\$CBB1,R14\$CBB2,R14\$CBB3,R14\$CBB4,R14\$CBB5,R14\$CBB6	R0402	270Ω	CRCW0402270RFKED	VISHAY(威世)	C482120	LCSC	CRCW0402270RFKED	270Ω
54	7	22Ω	R21\$CBB23,R22\$CBB23,R77,R96,R135,R149,R182\$CBB23	R0402	22Ω	0402WGF220JTCE	UNI-ROYAL(厚声)	C25092	LCSC	0402WGF220JTCE	22Ω
55	4	33Ω	R27\$CBB23,R28\$CBB23,R29\$CBB23,R30\$CBB23	R2512	33Ω	CRH2512F33R0E04Z	EVER OHMS(天二科技)	C175260	LCSC	CRH2512F33R0E04Z	33Ω
56	8	1.5kΩ	R32\$CBB16,R98\$CBB16,R100\$CBB16,R102\$CBB16,R104\$CBB16,R106\$CBB16,R134,R142	R0402	1.5kΩ	0402WGF1501TCE	UNI-ROYAL(厚声)	C25867	LCSC	0402WGF1501TCE	1.5kΩ

57	23	51kΩ	R32\$CBB17,R32\$CBB18,R32\$CBB20,R64\$CBB17,R64\$CBB18,R64\$CBB20,R96\$CBB17,R96\$CBB18,R96\$CBB20,R128\$CBB17,R128\$CBB18,R128\$CBB20,R130\$CBB17,R130\$CBB18,R130\$CBB20,R132\$CBB17,R132\$CBB18,R132\$CBB20,R134\$CBB17,R134\$CBB18,R134\$CBB20,R136\$CBB17,R136\$CBB20	R0402	51kΩ	0402WGF5102TCE	UNI-ROYAL(厚声)	C25794	LCSC	0402WGF5102TCE	51kΩ
58	8	0Ω	R45,R65,R148,R158\$CBB22,R158\$CBB23,R159\$CBB23,R164\$CBB22/CBB2,R174\$CBB22/CBB2	R0402	0Ω	0402WGF000TCE	UNI-ROYAL(厚声)	C17168	LCSC	0402WGF000TCE	0Ω
59	1	6.49kΩ	R46	R0402	6.49kΩ	RT0402BRE076K49L	YAGEO(国巨)	C853380	LCSC	RT0402BRE076K49L	6.49kΩ
60	7	1.8kΩ	R48\$CBB16,R99\$CBB16,R101\$CBB16,R103\$CBB16,R105\$CBB16,R107\$CBB16,R212\$CBB24	R0402	1.8kΩ	0402WGF1801TCE	UNI-ROYAL(厚声)	C25871	LCSC	0402WGF1801TCE	1.8kΩ
61	3	12kΩ	R51\$CBB23,R78,R183\$CB B22	R0402	12kΩ	0402WGF1202TCE	UNI-ROYAL(厚声)	C25752	LCSC	0402WGF1202TCE	12kΩ
62	3	15kΩ	R52,R136,R138	R0402	15kΩ	0402WGF1502TCE	UNI-ROYAL(厚声)	C25756	LCSC	0402WGF1502TCE	15kΩ
63	3	560kΩ	R53,R128,R140	R0402	560kΩ	0402WGF5603TCE	UNI-ROYAL(厚声)	C132339	LCSC	0402WGF5603TCE	560kΩ
64	1	8.2kΩ	R55	R0402	8.2kΩ	CR0402FF8201G	LIZ(丽智电子)	C100538	LCSC	CR0402FF8201G	8.2kΩ
65	3	40.2kΩ	R56,R129,R143	R0402	40.2kΩ	0402WGF4022TCE	UNI-ROYAL(厚声)	C25893	LCSC	0402WGF4022TCE	40.2kΩ
66	1	24kΩ	R57	R0402	24kΩ	0402WGF2402TCE	UNI-ROYAL(厚声)	C25769	LCSC	0402WGF2402TCE	24kΩ
67	4	0Ω	R58,R58\$CBB23,R132,R146	R1210	0Ω	RS-1210000JT	FH(风华)	C140000	LCSC	RS-1210000JT	0Ω
68	1	19.1kΩ	R59	R0402	19.1kΩ	0402WGF1912TCE	UNI-ROYAL(厚声)	C52036	LCSC	0402WGF1912TCE	19.1kΩ
69	2	36kΩ	R60,R131	R0402	36kΩ	0402WGFJ0363TCE	UNI-ROYAL(厚声)	C25556	LCSC	0402WGFJ0363TCE	36kΩ
70	2	27kΩ	R80\$CBB16,R97\$CBB16	R0402	27kΩ	0402WGF2702TCE	UNI-ROYAL(厚声)	C25771	LCSC	0402WGF2702TCE	27kΩ
71	1	43kΩ	R125	R0402	43kΩ	0402WGFJ0433TCE	UNI-ROYAL(厚声)	C25561	LCSC	0402WGFJ0433TCE	43kΩ
72	2	82kΩ	R126,R147	R0402	82kΩ	0402WGF8202TCE	UNI-ROYAL(厚声)	C4142	LCSC	0402WGF8202TCE	82kΩ
73	2	12kΩ	R127,R219\$CBB24	R0402	12kΩ	0402WGFJ0123TCE	UNI-ROYAL(厚声)	C25535	LCSC	0402WGFJ0123TCE	12kΩ
74	1	32.4kΩ	R133	R0402	32.4kΩ	0402WGF3242TCE	UNI-ROYAL(厚声)	C26974	LCSC	0402WGF3242TCE	32.4kΩ

75	4	68kΩ	R141,R144,R150,R151	R0402	68kΩ	0402WGF68 02TCE	UNI-ROYAL(厚声)	C36871	LCSC	0402WGF68 02TCE	68kΩ
76	1	75kΩ	R145	R0402	75kΩ	0402WGF75 02TCE	UNI-ROYAL(厚声)	C25798	LCSC	0402WGF75 02TCE	75kΩ
77	2	10Ω	R215\$CBB24,R217\$CBB24	R0402	10Ω	0402WGF10 0JTCE	UNI-ROYAL(厚声)	C25077	LCSC	0402WGF10 0JTCE	10Ω
78	9	Test-Point	TP1,TP2,TP2\$CBB23,TP3, TP3\$CBB23,TP4,TP5,TP6, TP7	Test-Point-0. 5mm						Test-Point	
79	32	MCP6024T-I/ SL-MS	U1\$CBB1,U1\$CBB2,U1\$C BB3,U1\$CBB4,U1\$CBB5,U 1\$CBB6,U1\$CBB8,U1\$CBB 10,U1\$CBB11,U1\$CBB12, U1\$CBB13,U1\$CBB14,U1\$ CBB15,U1\$CBB16,U1\$CBB 17,U1\$CBB18,U1\$CBB20, U2\$CBB15,U2\$CBB16,U2\$ CBB17,U2\$CBB18,U2\$CBB 20,U3\$CBB9,U3\$CB515,U 3\$CBB19,U3\$CBB49,U4\$C BB9,U4\$CBB19,U4\$CBB49 ,U5\$CBB9,U5\$CBB19,U5\$ CBB49	SOIC-14 L8. 7-W3.9-P1.2 7-LS6.0-BL-1		MCP6024T-I/ SL-MS	MSKSEMI(美 森科)	C54539958	LCSC	MCP6024T-I/ SL-MS	MCP6024T-I/ SL-MS
80	12	XC2361A	U1\$CBB7,U1\$CBB21,U2\$C BB7,U2\$CBB21,U3\$CBB7, U3\$CBB21,U4\$CBB7,U4\$C BB21,U5\$CBB7,U5\$CBB21 ,U7\$CBB7,U7\$CBB21	TSOT-23-6 L2.9-W1.6-P 0.95-LS2.6-B L		XC2361A	SiChange(芯 昌科技)	C41383383	LCSC	XC2361A	XC2361A
81	1	T20F256C3	U1\$CBB23	FBGA-256 L 13.0-W13.0- R16-C16-P0. 80-TL		T20F256C3	Efinix	C3277960	LCSC	T20F256C3	T20F256C3
82	11	PZ2.54-2*4	U1,U1\$CBB22,U1\$CBB24, U2,U2\$CBB24,U9\$CBB23 ,U9\$CBB24,U110\$CBB24, U111\$CBB24,U112\$CBB22 ,U112\$CBB24	HDR-TH 8P- P2.54-V-M-R 2-C4-S2.54		PZ2.54-2*4	ZHOURI(洲 白)	C5156676	LCSC	PZ2.54-2*4	PZ2.54-2*4
83	2	PCA9306DC 1	U2\$CBB22/CBB1,U95\$CBB 22/CBB1	VSSOP-8 L2 .4-W2.1-P0.5 0-LS3.2-BR		PCA9306DC 1	NXP(恩智浦)	C129510	LCSC	PCA9306DC 1	PCA9306DC 1
84	1	KF301-5.0-2 P	U3	CONN-TH P 5.00 KF301- 5.0-2P		KF301-5.0-2 P	KEFA(科发)	C474881	LCSC	KF301-5.0-2 P	KF301-5.0-2 P
85	3	AP64350SP- 13	U5,U6,U7	SO-8 L4.9- W3.9-P1.27- LS6.0-BL-EP		AP64350SP- 13	DIODES(美 台)	C2071691	LCSC	AP64350SP- 13	AP64350SP- 13

86	30	74HC595	U6\$CBB16/CBB1,U6\$CBB16/CBB2,U6\$CBB16/CBB3,U6\$CBB16/CBB4,U6\$CBB16/CBB5,U6\$CBB16/CBB6,U6\$CBB17/CBB1,U6\$CBB17/CBB2,U6\$CBB17/CBB3,U6\$CBB17/CBB4,U6\$CBB17/CBB5,U6\$CBB17/CBB6,U6\$CBB17/CBB7,U6\$CBB17/CBB8,U6\$CBB18/CBB1,U6\$CBB18/CBB2,U6\$CBB18/CBB3,U6\$CBB18/CBB4,U6\$CBB18/CBB5,U6\$CBB18/CBB6,U6\$CBB18/CBB7,U6\$CBB18/CBB8,U6\$CBB20/CBB1,U6\$CBB20/CBB2,U6\$CBB20/CBB3,U6\$CBB20/CBB4,U6\$CBB20/CBB5,U6\$CBB20/CBB6,U6\$CBB20/CBB7,U6\$CBB20/CBB8	SOP-16 L9.9-W3.9-P1.27-LS6.0-BL		74HC595	GR(国睿)	C18164493	LCSC	74HC595	74HC595
87	3	MP2393GTL-Z	U9,U12,U16	SOT-583-8 L2.1-W1.2-P0.50-LS1.6-BL		MP2393GTL-Z	MPS(芯源)	C6674609	LCSC	MP2393GTL-Z	MP2393GTL-Z
88	2	APX803S-31 SA-7	U14,U118\$CBB23	SOT-23-3 L2.9-W1.3-P1.9 0-LS2.4-BR		APX803S-31 SA-7	DIODES(美台)	C129757	LCSC	APX803S-31 SA-7	APX803S-31 SA-7
89	1	W25Q64JVZ PIQ	U19\$CBB23	WSON-8 L6.0-W5.0-P1.2 7-BL-EP		W25Q64JVZ PIQ	WINBOND(华邦)	C2940197	LCSC	W25Q64JVZ PIQ	W25Q64JVZ PIQ
90	1	RTL8211F-C G	U98\$CBB22/CBB2	WQFN-40 L5.0-W5.0-P0.4-BL-EP		RTL8211F-C G	REALTEK(瑞昱)	C187932	LCSC	RTL8211F-C G	RTL8211F-C G
91	1	FT2232HL-R EEL	U98\$CBB24	LQFP-64 L1 0.0-W10.0-P0.50-LS12.0-TL		FT2232HL-R EEL	FTDI(飞特帝亚)	C27882	LCSC	FT2232HL-R EEL	FT2232HL-R EEL
92	1	W25Q128JV SIQ	U99\$CBB22	SOIC-8 L5.3-W5.3-P1.27-LS8.0-BL		W25Q128JV SIQ	Winbond(华邦)	C97521	LCSC	W25Q128JV SIQ	W25Q128JV SIQ
93	1	T120F324C3	U100\$CBB22	BGA324C65 P18X18 120 0X1200X97N						T120F324C3	T120F324C3
94	1	12MHz	U101\$CBB24	CRYSTAL-S MD 4P-L3.2-W2.5-BL	12MHz	ECS-120-10-33B-CKM-TR	ECS	C2450718	LCSC	ECS-120-10-33B-CKM-TR	12MHz
95	2	BLM15PX33 0SN1D	U103\$CBB24,U104\$CBB24	L0402		BLM15PX33 0SN1D	muRata(村田)	C160974	LCSC	BLM15PX33 0SN1D	BLM15PX33 0SN1D
96	1	USB-B10-BRW	USB3\$CBB24	USB-B-TH USB-B10-BRW		USB-B10-BRW	XUNPU(讯普)	C720549	LCSC	USB-B10-BRW	USB-B10-BRW
97	2	50MHz	X1,X1\$CBB23	OSC-SMD 4 P-L3.2-W2.5-BL	50MHz	OT322550MJB A4SL	YXC(扬兴晶振)	C717684	LCSC	OT322550MJB A4SL	50MHz

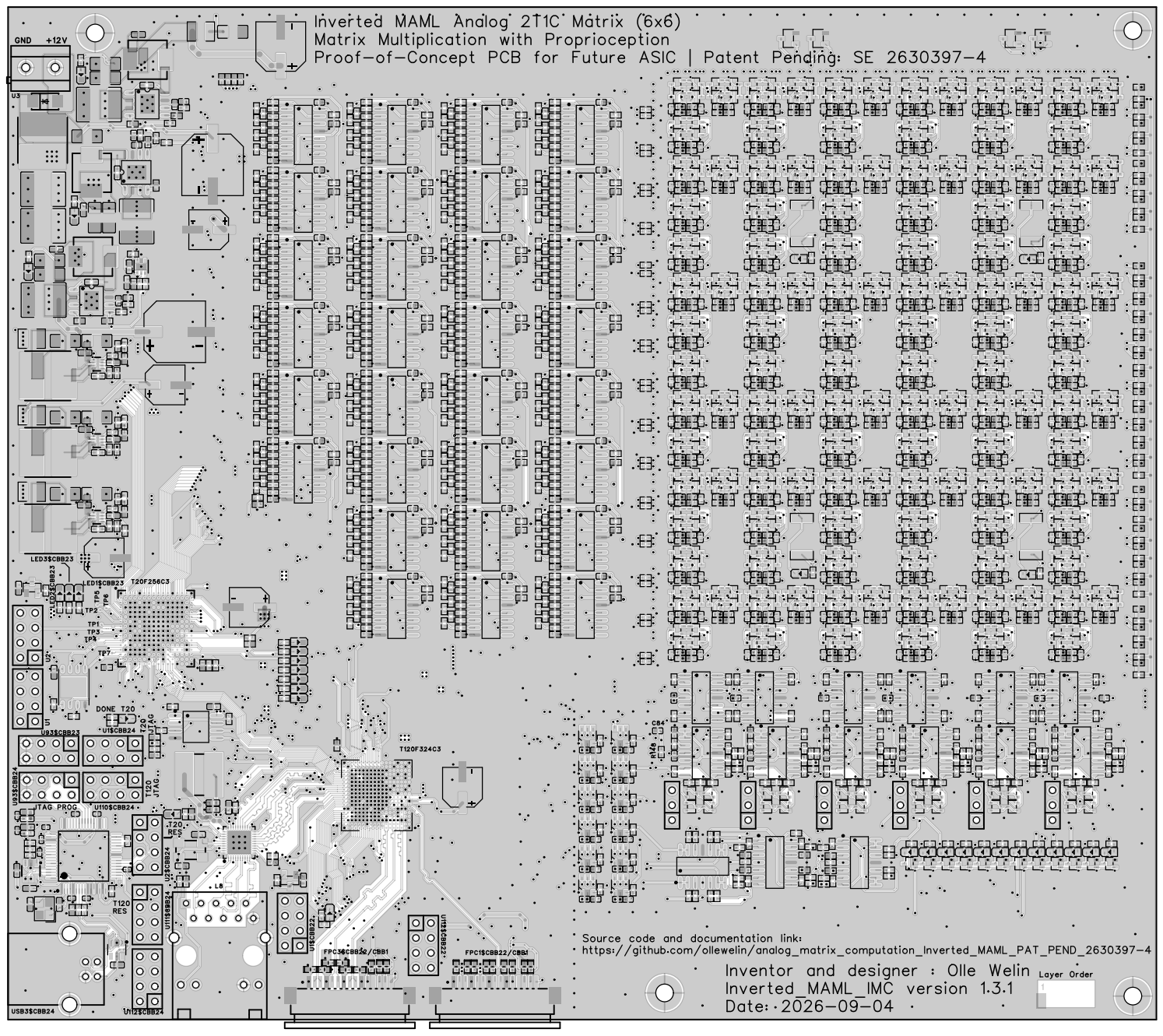
98	1	25MHz	X4\$CBB22/CBB2	CRYSTAL-S MD_4P-L5.0- W3.2-BL	25MHz	XC53M4-25. 000-F12JJDT	HCI(杭晶)	C2916916	LCSC	XC53M4-25. 000-F12JJDT	25MHz
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[illegible]

3A					
					30V

Inverted MAML Analog 2T1C Matrix (6x6)
Matrix Multiplication with Proprioception

Proof-of-Concept PCB for Future ASIC | Patent Pending: SE 2630397-4



Source code and documentation link:

Source code and documentation link:
https://github.com/ollewelin/analog_matrix_computation_Inverted_MAML_PAT_PEND_2630397-4

Inventor and designer : Olle Welin

Inverted MAML IMC version 1.3.1

Date: 2026-09-04

Layer Order

